

L1.8 Linearization & Taylor

Recall: L1.7b Related Rates

Name _____

Solo – 9 minutes

1. Differentiate each with respect to t .

(a) $A = \pi r^2$

(b) $V = \frac{1}{3}\pi r^2 h$

(c) One of these will stall when you try to solve for a rate. Which one, and **how could you tell from the MEQ**, before differentiating?

2. Oil spreads in a circle on still water. The radius grows at 0.5 m/min. How fast is the area growing when the radius is 20 m?

Use the template. You already have the REQ from 1(a).

3. Now the slick has stopped spreading. Its radius is not changing at all — it is simply **measured**, as 20 ± 0.1 m.

What can you say about its area? Write down what you think the calculation should be.